



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam: Spring 2026

Course Code: CSE 3811, Course Title: Artificial Intelligence

Total Marks: 40

Duration: 2 hours

Answer all questions. Marks are indicated on the right side of each question.

[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

1. A university department needs to schedule five courses into four available exam time slots: Time Slot 1 (T1), Time Slot 2 (T2), Time Slot 3 (T3), and Time Slot 4 (T4). [5]

The courses are: Data Structures (DS), Algorithms (AL), Database Systems (DB), Artificial Intelligence (AI), and Computer Networks (CN).

Each course must be assigned exactly one time slot. The following constraints must be satisfied:

- Some students are enrolled in both **Data Structures** and **Algorithms**, so these two courses cannot be scheduled in the same time slot.
- **Algorithms** and **Artificial Intelligence** share the same instructor; therefore, they cannot be scheduled in the same time slot.
- **Computer Networks** must be scheduled before **Database Systems**.
- **Artificial Intelligence** cannot be scheduled in **Time Slot 1** because the instructor is unavailable.
- **Computer Networks** can only be scheduled in **Time Slot 3** or **Time Slot 4**.
- **Data Structures** and **Database Systems** must be scheduled on the same day. Assume Time Slots T1 and T2 belong to Day 1, and T3 and T4 belong to Day 2.
- At most **two courses** can be assigned to any single time slot.
- If **Algorithms** is scheduled in **Time Slot 2**, then **Computer Networks** must be scheduled in **Time Slot 4**.

Now, formulate this situation as a Constraint Satisfaction Problem (CSP) and determine a valid examination schedule for all courses. You must solve the CSP using MRV and LCV heuristics.

2. (a) A university has three computer laboratories: **Lab A**, **Lab B**, and **Lab C**. [3]

- In **Lab A**, 8 out of 20 computers have SSD storage.
- In **Lab B**, 12 out of 30 computers have SSD storage.
- In **Lab C**, 15 out of 25 computers have SSD storage.

A laboratory is selected randomly (each laboratory has an equal probability of being selected), and then a computer is chosen at random from that laboratory. If the selected computer is found to have **SSD storage**, what is the probability that it came from **Lab C**?

(b) A survey was conducted among **220 football fans** to determine their favorite FIFA World Cup team. Among the surveyed fans:

- Brazil was the favorite team of 90 fans, of whom 60 were male and 30 were female.
- Argentina was the favorite team of 80 fans, of whom 75% were male.
- France was the favorite team of the remaining fans, of whom 20% were female.

Let **T** denote the favorite team and **gN** denote gender.

- i. Construct the joint probability distribution table $P(T, gN)$ for the variables **T (Team)** and **gN (Gender)**. [2]
- ii. If a fan's favorite team is France, what is the probability that the fan is female? [2]

3. A company has developed a **smart home security system** that automatically monitors unusual activities and sends alerts to homeowners. The system considers five factors:

1. **Burglary (B)**
2. **Power Failure (P)**
3. **Alarm (A)**
4. **Mobile Notification (M)**
5. **Security Guard Call (G)**

Historical records show that a burglary occurs in about **2%** of cases, while a power failure occurs in about **5%** of cases.

The alarm system is influenced by both burglary and power failure. When both a burglary and power failure occur, the alarm rings with probability **0.98**. When a burglary occurs without a power failure, the alarm rings with probability **0.95**. When there is no burglary but a power failure occurs, the alarm rings with probability **0.40** due to electrical disturbances. When neither event occurs, the alarm may still ring falsely with probability **0.01**.

If the alarm rings, the smart system successfully sends a mobile notification to the homeowner with probability **0.90**. If the alarm does not ring, a notification may still be sent due to a system error with probability **0.10**.

Similarly, if the alarm rings, an automatic call is placed to a security guard with probability **0.85**. If the alarm does not ring, such a call is mistakenly generated with probability **0.05**.

Please answer the following questions strictly following the given scenario.

- a. Construct the **Bayesian Network** representing the relationships among the five variables and specify the corresponding **Conditional Probability Tables (CPTs)**. [3]
 - b. Using the Bayesian Network, **calculate the probability** that a mobile notification is received given that a burglary has occurred [4]
4. The organizers of the World Martial Arts Tournament want to develop a classifier that predicts whether a fighter is likely to Win or Lose a match based on their Race, Power Level, and Transformation Status.

ID	Race	Power Level	Transformation	Class
1	Saiyan	High	Yes	Win
2	Saiyan	Medium	No	Win
3	Namekian	Medium	No	Win
4	Human	Low	No	Lose
5	Saiyan	High	No	Win
6	Human	Medium	Yes	Lose
7	Namekian	Low	No	Lose
8	Saiyan	Medium	Yes	Win
9	Human	Medium	No	Lose
10	Namekian	High	Yes	Win

- a. For a new fighter with the following attributes **Race = Namekian, Power Level = Medium, Transformation = Yes**, use the Naïve Bayes Classifier with **Laplacian Smoothing (k = 2)** to determine whether the fighter is more likely to Win or Lose the match. Show all calculations. [6]
- b. Suppose no **winning fighter** in the training data is a **Namekian**. How does **Laplacian smoothing** help when classifying a new **Namekian fighter**? [1]

5. You are building a **Decision Tree** model to predict concert outcomes based on some popular Japanese artists' attributes.

Singer	Community	Merchandise Purchase	Genre	Concert Outcome
Ado	Active	Yes	Ballad	Success
Ado	Active	Yes	Rock	Success
milet	Active	Yes	Ballad	Success
milet	Inactive	No	Rock	Failure
YUI	Active	No	Rock	Failure
YUI	Inactive	No	Rock	Failure
milet	Active	No	Ballad	Success
Vaundy	Inactive	Yes	Ballad	Success
milet	Inactive	Yes	Rock	Failure
YUI	Active	Yes	Ballad	Success

- a. What will be the **root feature** of the Decision Tree model? [6]
- b. For the next **Ballad** concert of Ado, the community is active, with good merchandise sales, predict if the concert will be a success or a failure. [1]

6. A sports analytics firm is studying how football fans switch their primary team allegiance among three powerhouse nations: Brazil, Argentina, and France. They surveyed fans on their favourite team to win at the 2022 World Cup and tracked how allegiances changed by the 2026 World Cup.

Initial (**2022 World Cup**) survey results (**total of 200 fans**):

- 120 primarily supported **Brazil**
- 50 supported **Argentina**
- 30 supported **France**

By the **2026 World Cup**, the following shifts were observed:

- Among the 120 **Brazil** fans in 2022, 24 switched their allegiance to **Argentina**, 12 switched to **France**, and the rest remained **Brazil supporters**.

- Among the **50 Argentina** fans in **2022**, **15** switched to **Brazil**, **5** switched to **France**, and the **rest** stayed with **Argentina**.
- Among the **30 France** fans in **2022**, **9** switched to **Brazil**, **3** switched to **Argentina**, and the **rest** kept supporting **France**.

Assume these transition patterns remain the same for each **subsequent World Cup cycle** (**2026→2030**, **2030→2034**, **2034→2038**, and so on).

- Construct a **transition matrix** representing the allegiance changes from any World Cup t to the next World Cup $t+1$. [2]
- Using the 2022 survey, predict the percentage of fans who will support **Argentina** as their primary team at the **2034 World Cup**. [3]
- Calculate the **stationary distribution** that a randomly selected fan supports each of the three teams. [2]